Eval2
Cohort 3 Group 5

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User Evaluation Report

Brownfield Development

Methods of Data Elicitation and Processing

During the user evaluation phase of the project, data was collected through the recruitment of test users to provide the development team with issues that could be raised and appropriate feedback to aid in the development of the game and remedy any issues provided by the users.

In order to recruit members for the user evaluation research, students who had experience in software development were approached informally and were asked to participate in a prototype testing. These participants were fully informed and were provided with informed consent forms that gave a complete insight into what the project was, and how their feedback would have been used, in accordance with the University of York's ethical guidelines on data elicitation and procedure.

The evaluation followed a task-oriented observational approach chosen because it allows usability issues to be identified during natural usage of the game, providing a free environment for the participants to interact and find potential issues within the system. A mixed observation approach was used to allow for participants to complete tasks alongside the team members or were emailed and asked to complete their tasks when they could.

Each participant was provided with a short questionnaire which allowed them to express any issues they may have found within the game, providing a severity rating of how much it impacted their experience. Each participant was asked to complete the following tasks:

- Start and move around the game's arena.
- Interact with in-game interactable objects and obstacles.
- Complete the game's primary objective.

and following this were issued their questionnaires.

The combination of qualitative observation and quantitative severity ratings improves the reliability and usefulness of evaluation results, particularly for early-stage prototypes[1]. Overall, this evaluation method was appropriate for the project's scope, maturity, and interactive nature.

Identification of Usability Problems

The table below summarises the main usability issues raised during the user evaluation phase of the game, following severity ratings provided by users (1-5, where 1 indicates a minor issue and 5 indicates a severe usability issue.)

Problem ID	**Usability Problem Description**	**Affected User(s)** (Severity Rating >= 2)	**Average Severity Rating (1-5)**
P1	**Gameplay/Technical Issues:** Significant bugs or glitches encountered during movement and interaction.	U3, U4	**2.86**
P2	**UX Information:** Users found the on-screen information/instructions insufficient or confusing.	U7, U6, U3	**3.14**
P3	**UX Usability:** General difficulty in navigating the interface or understanding the game flow.	U3, U4, U5, U6, U7	**3.57**
P4	**UX Controls:** Lack of intuitiveness in the control scheme or mapped inputs.	U1, U2, U3, U4, U5, U6, U7	**3.71**
P5	**Objective Clarity:** Minor confusion regarding the specific goals or "what to do next."	U1, U2, U3, U4, U5, U6, U7	**3.57**
P6	**System Stability:** Occasional frame drops or performance lag during arena movement.	U7	**0.86**

Bibliography

[1] H. Sharp, Y. Rogers, and J. Preece, Interaction Design: Beyond Human–Computer Interaction, 6th ed. Hoboken, NJ, USA: Wiley, 2023, ch. 15, “Evaluation studies: from controlled to natural settings.”